SBEEH Software

INSTRUCTOR-LED COURSE · 3 DAYS · 6 HOURS A DAY

DESIGN AND MANAGE ANALYTICS SOLUTIONS USING POWER BI



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Microsoft CERTIFIED TRAINER MCT

ABOUT ME

MOHAMMAD SBEEH

Managing & Technical Director · Microsoft Certified Trainer (MCT)

20+ years as a solutions architect and technical lead in software development

FOUNDER	SBEEH Software · www.sbeeh.com
CERTIFIED	MCP, MCTS, MCPD, MCITP, MCT and C# Corner MVP
FOLLOW	Linkedin.com/in/sbeeh

Digital Transformation · Software Consulting · Training · Outsourcing

YOUR TURN

STUDENT INTRODUCTIONS

01 Name and role

02 Company or affiliation

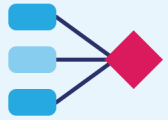
03 Experience with data analytics and Power BI

04 Expectations for the course



HANDS-ON COURSE · 18 HOURS

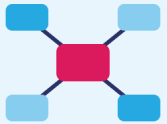
FROM ZERO TO DASHBOARD IN THREE DAYS



DAY 1

CONNECT & PREPARE

- Data analytics and Power BI
- Training environment (Lab 0)
- Get data: files, SQL, API, web
- Storage modes
- Data profiling



DAY 2

TRANSFORM & MODEL

- Power Query Editor
- Append and Unpivot
- Contoso sales case study
- Relationships and the data model
- Main lab: build the model



DAY 3

ANALYZE, VISUALIZE & MANAGE

- DAX measures and calculated columns
- Report design and layout
- Publishing to the service
- Row-level security

LOGISTICS

SIX HOURS A DAY, THREE SESSIONS



Block	Time	Length	What happens
Session 1	9:00 – 10:45 AM	105 min	Concepts, demos and first labs
Break 1	10:45 – 11:00 AM	15 min	Short break
Session 2	11:00 AM – 1:00 PM	120 min	Guided labs
Break 2	1:00 – 2:00 PM	60 min	Long break
Session 3	2:00 – 3:00 PM	60 min	Main lab, recap and questions

10 MINUTES ASSESSMENT

PRE-ASSESSMENT

10

minutes

A quick check of what you already know, before we start.



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INTRODUCTION

DATA ANALYTICS

THE DATA

GENERATED EVERY DAY

- Transactional databases such as SQL or Oracle
- Services, like REST APIs
- Other areas, like social media

THE CHALLENGES

TELLING AN ACCURATE STORY

- Different sources and structures, at the same time
- Relationships between all that data
- Reports and dashboards without depending on IT

THE ANSWER

POWER BI

- Raw data into insights, through interactive dashboards and reports
- Real-time analysis across many data sources
- User-friendly, mobile and secure

Part of Microsoft Power Platform: a suite of tools to automate processes, analyze data and build custom applications without deep technical expertise.

FROM DATA TO DECISIONS



An accurate story, so decision-makers make better business decisions

THE POWER BI ROADMAP

FIVE THINGS EVERY DATA ANALYST DOES

01
PREPARE
Clean and transform data so it is ready to model and visualize.

Days 1–2

02
MODEL
Build relationships between the tables.

Day 2

03
VISUALIZE
Create a graphical representation of the data.

Day 3

04
ANALYZE
Uncover insights with DAX.

Day 3

05
MANAGE
Manage your Power BI assets.

Day 3

From raw data to real insight

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DAY 1 OF 3 · CHAPTER 01

CONNECT & PREPARE

Import from Excel, SQL and the web, then get to know the data before you shape it.

105 min	Welcome, data analytics, Power BI and Lab 0
15 min	Break
120 min	Getting data, storage modes, Labs 01–04
60 min	Break
60 min	Data profiling (Lab 05) and recap

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DAY 1 · INTRODUCTION

GETTING STARTED WITH DATA ANALYTICS

Decision-makers depend on an accurate story to drive better business decisions.

The challenge: bring data from different sources, with different structures, together into one reliable picture.

- **Transactional data** in databases such as SQL Server or Oracle
- **Services** exposed through REST APIs
- **Other areas** such as social media and the web

DAY 1 · INTRODUCTION TO POWER BI

WHAT IS POWER BI?

A collection of software, services, apps and connectors that work together to turn unrelated sources of data into visual, interactive insights.

Power BI Desktop

1 · BUILD

POWER BI DESKTOP

A free application on your computer to connect to, transform and visualize data.

2 · PUBLISH AND SHARE

POWER BI SERVICE

The online service where reports are published, dashboards are built and content is shared.



3 · VIEW

POWER BI MOBILE

Apps for phones and tablets to view and interact with shared reports and dashboards.

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THE COMMON FLOW OF WORK



1 · DESKTOP

GET DATA

Data is cleaned, transformed and modeled, and reports are created.



2 · SERVICE

PUBLISH

Reports go to the Power BI service, where you build dashboards or new visualizations.



3 · SERVICE

SHARE

Share dashboards and other Power BI assets with others.



4 · MOBILE / WEB

VIEW

View and interact with shared reports in the mobile apps or a browser.

LAB 0 · HANDS-ON

PREPARING THE TRAINING ENVIRONMENT

01 Install Power BI Desktop

https://powerbi.microsoft.com/en-us/downloads/

02 Create a free Office 365 account

https://www.microsoft.com/en-us/microsoft-365/enterprise/office-365-e5

03 Download the lab files

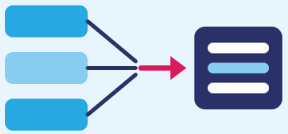
https://github.com/sbeeh-software/BI

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DAY 1 · GET DATA IN POWER BI

INTRODUCTION TO GETTING DATA

The first step in the data analysis process is identifying and getting data. It usually sits in different databases and files, yet all of it is related.



EXTRACT

Pull data from the various sources: SQL Server, Excel, services and more.



CLEAN

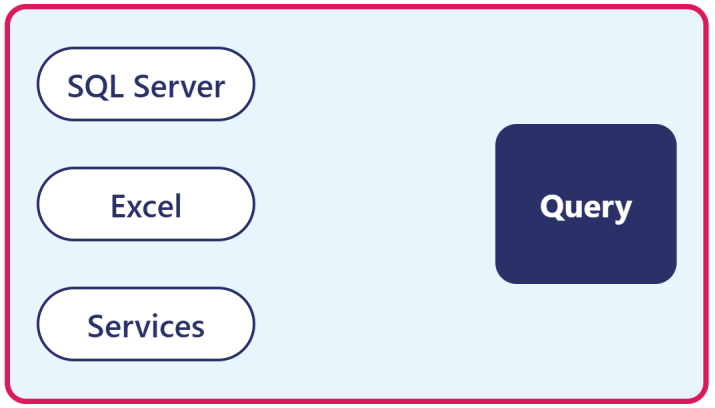
Use Power Query to rename columns, replace values, remove errors and combine query results.



LOAD

Load the cleaned data into Power BI, ready to model and report on.

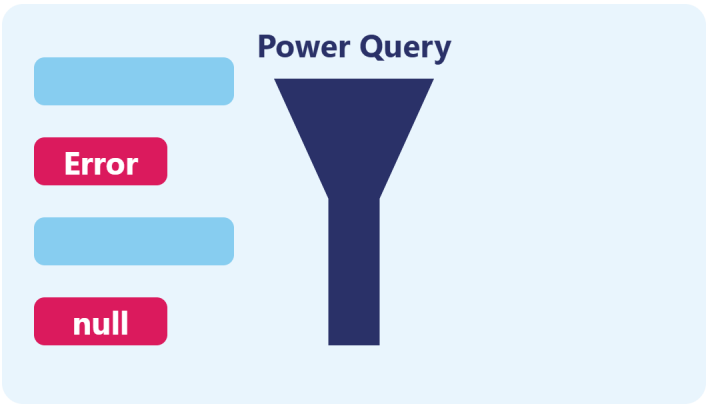
EXTRACT, CLEAN, LOAD IN MOTION



01 · EXTRACT

Pull from the sources

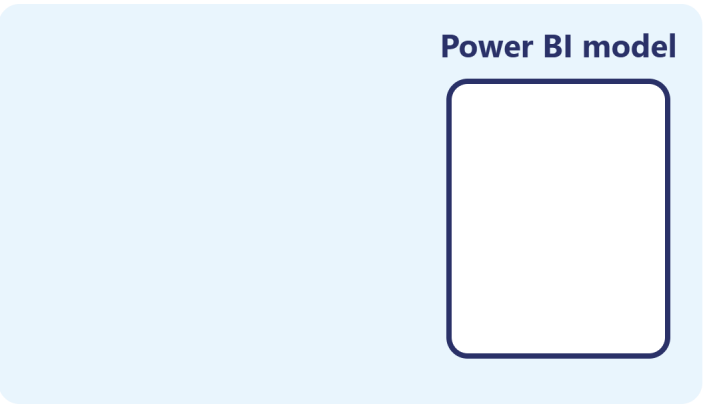
SQL Server, Excel, services and more.



02 · CLEAN

Shape it with Power Query

Rename, replace, remove errors, combine.



03 · LOAD

Load into Power BI

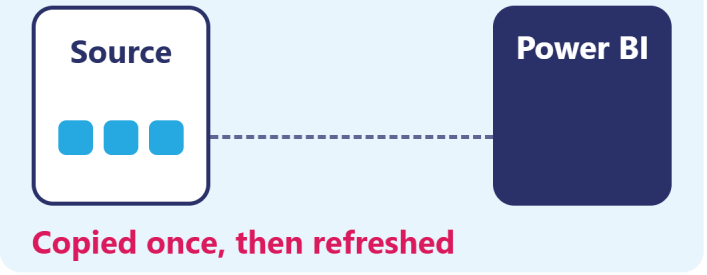
Ready to model and report on.

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DAY 1 · GET DATA IN POWER BI

STORAGE MODE

Where does the data live? It decides the **available transformations** and **report performance**.



Copied once, then refreshed

PREFERRED

Import

A copy lives in Power BI. Fast, and supports all transformations.

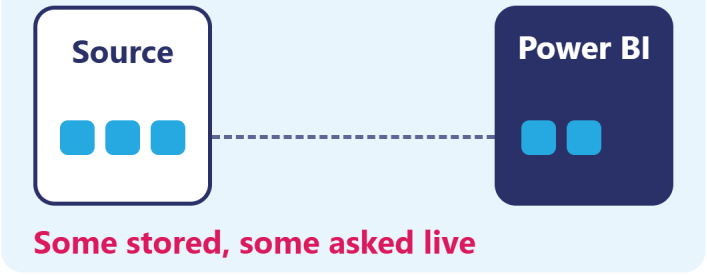


Asks the source every time

LIVE QUERIES

DirectQuery

Data stays in the source. Always current, fewer transformations.



Some stored, some asked live

MIXED

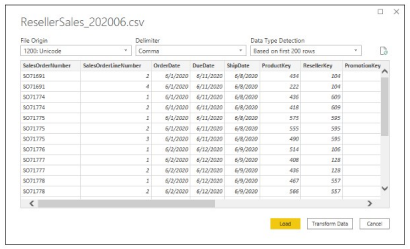
Dual (Composite)

Some tables imported, some queried directly, in one model.

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LABS 01–04 · HANDS-ON

GET DATA FROM FOUR KINDS OF SOURCE



LAB 01

FLAT FILES

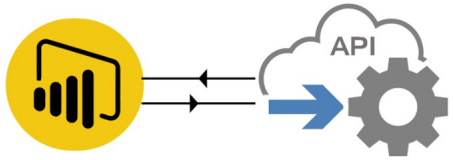
One table, every row in the same structure: CSV, Excel or text. We load an Excel sheet.



LAB 02

RELATIONAL DATA SOURCES

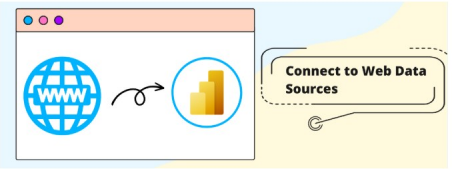
Connect to a database in the cloud or on-premises. We connect to Azure SQL Server.



LAB 03

SERVICE / API

Power BI supports connecting to APIs. We read data from a REST API.



LAB 04

WEBSITE

Connect to data published on a public web page.

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DAY 1 · LAB 05

DATA PROFILING



Data profiling is the process of understanding more about the data. Power Query Editor offers five view options:

MONOSPACED

Fixed-width font for the preview.

SHOW WHITESPACE

Reveals spaces and line breaks.

COLUMN QUALITY

Share of valid, error and empty values.

COLUMN DISTRIBUTION

Distinct and unique value counts.

COLUMN PROFILE

Statistics and value frequency.

END OF DAY 1

DAY 1 RECAP

Tomorrow: transform the data and build the model.

- ✓ The five tasks of a data analyst
- ✓ Power BI Desktop, service and mobile
- ✓ Files, SQL, API and web sources
- ✓ Storage modes and data profiling

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DAY 2 OF 3 · CHAPTERS 01–03

TRANSFORM & MODEL

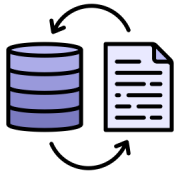
Clean data with Power Query, then build relationships and a solid data model.

105 min	Power Query Editor, common transformations, Labs 06–07
15 min	Break
120 min	Main lab: the Contoso sales report
60 min	Break
60 min	Data modeling and recap

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DAY 2 · CLEAN AND TRANSFORM THE DATA

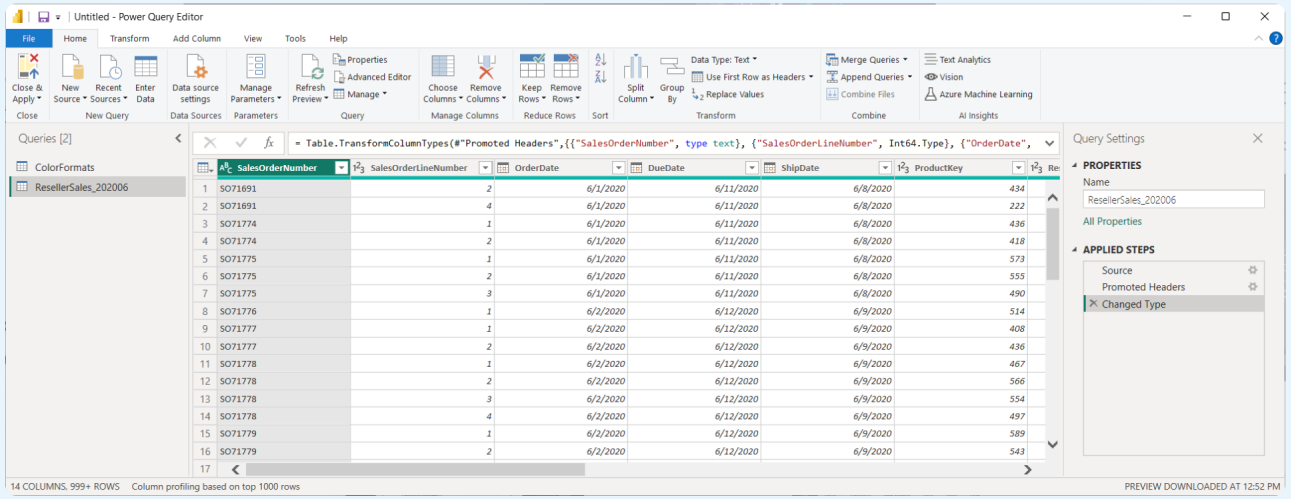
DATA TRANSFORMATION



Converting data from one format, structure or value set into another.

So it can be used for analysis and reporting.

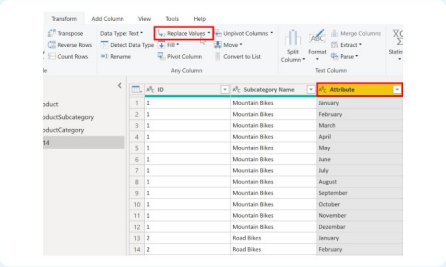
- Open the editor with **Transform data** on the Home tab
- Every shaping step is recorded
- Steps re-apply on each refresh
- Queries pane left, Query settings right



Power Query Editor: queries on the left, the data preview in the middle, and the applied steps on the right.

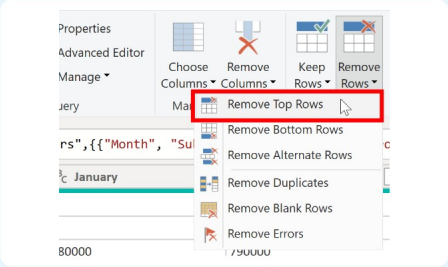
DAY 2 · CLEAN AND TRANSFORM THE DATA

COMMON TRANSFORMATIONS



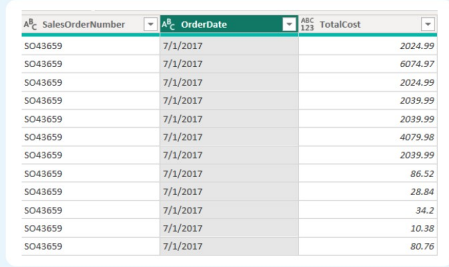
USER-FRIENDLY VALUES

Rename queries and columns so names are consistent and meaningful to the user.



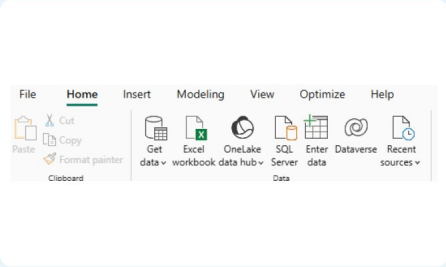
SHAPING TABLE STRUCTURE

Promote headers, remove rows and columns, split, extract and add new columns.



COLUMN DATA TYPES

Power BI detects types from the first 1,000 rows. Check them before loading: wrong types hurt performance.



ENTER DATA

Type a small static table straight into Power BI: lookup tables, parameters and the like.

LABS 06–07 · HANDS-ON

POWER QUERY: APPEND AND UNPIVOT

LAB 06

APPEND

Combines two or more tables by stacking them on top of each other.

The result holds all rows from the original tables in one unified table.

LAB 07

UNPIVOT

Converts columns into rows: from a wide format to a long format.

Used when the current structure is not ideal for visuals or calculations.

DAY 2 · MAIN LAB · SALES REPORT

CONTOSO CASE STUDY

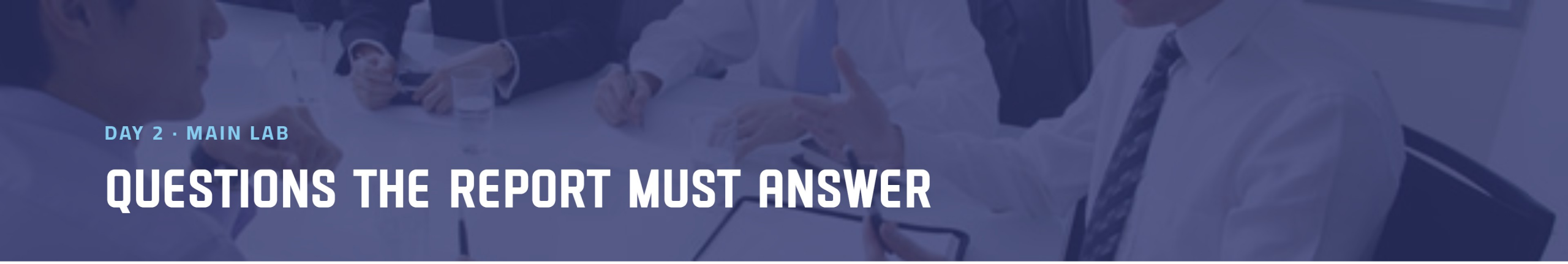
CASE STUDY

Contoso's sales team records every project it works on in one Excel sheet.

We turn that sheet into a sales report, and keep building on it for the rest of the course.



Contract Description	Status	Client Name	From	To	Total Amount
Project title	Open or Closed	Name of the client	Project start date	Project end date	Project amount



DAY 2 · MAIN LAB

QUESTIONS THE REPORT MUST ANSWER

INCOME OVER TIME

- Total income for all years
- Income across the years
- Income across years and months
- Months with no income recorded

HIGHEST AND LOWEST

- Year with the highest total income
- Year with the lowest total income
- Highest month in the selected year
- Lowest month in the selected year

CLIENTS AND COUNTRIES

- Income across the clients
- Top client by total income
- Top country by total income

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MAIN LAB · HANDS-ON

ADDITIONAL CHALLENGES

01 Show the full country name instead of the ISO code.

02 Use the full month name instead of the month number.

03 Display all months, whether they contain data or not.

04 For months without data, show the total amount as zero.

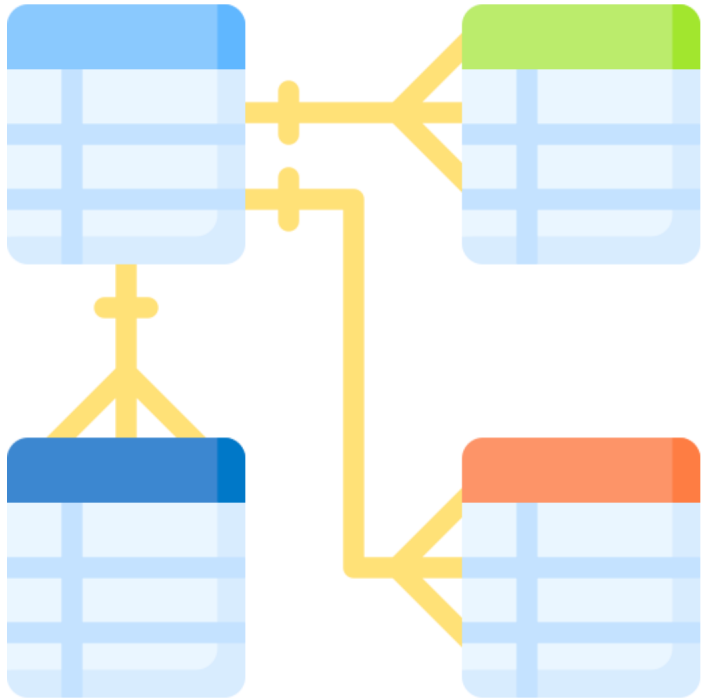


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INTRODUCTION TO DATA MODELING

Creating logical relationships between tables of data, to establish a structure that supports meaningful analysis and reporting.

- **ACCURATE REPORTS**
- **FASTER DATA EXPLORATION**
- **SIMPLER AGGREGATIONS**
- **EASIER TO MAINTAIN**



FIVE STEPS TO A DATA MODEL

01
IMPORT DATA
Load from Excel, databases or APIs.

02
CLEAN AND TRANSFORM
Power Query: remove duplicates, filter rows.

03
DEFINE RELATIONSHIPS
In Model view, drag fields to connect tables on common keys.

04
CONFIGURE THEM
Set the relationship properties.

05
VALIDATE THE MODEL
Test relationships and calculations.

MAIN LAB · HANDS-ON

DATA MODELING IN THE SALES REPORT

Implement data modeling in the Contoso sales report so it can answer the additional challenges.

01

Add lookup tables

Countries with full names, and a complete list of months.

02

Relate them to sales

Connect on common keys in Model view and set the properties.

03

Validate

Check that every month and every country name shows up.

END OF DAY 2

DAY 2 RECAP

Tomorrow: DAX, report design, publishing and security.

- ✓ Power Query Editor and applied steps
- ✓ Append, Unpivot and data types
- ✓ The Contoso sales case study
- ✓ Relationships and the data model

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DAY 3 OF 3 · CHAPTERS 02, 04, 05

ANALYZE, VISUALIZE & MANAGE

DAX formulas, interactive reports, and publishing to the Power BI service.

105 min	DAX: measures, calculated columns and the measures lab
15 min	Break
120 min	Main lab: create and design the report
60 min	Break
60 min	Publishing, row-level security and wrap-up

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INTRODUCTION TO DAX

A formula language for custom calculations in Power BI.

Similar to Excel formulas, but more powerful for working with relational data models.

CALCULATED COLUMNS

Create new columns based on existing data.

MEASURES

Define aggregations like SUM, AVERAGE or COUNT.

TWO FIRST FORMULAS

Basic aggregation

```
Total Sales = SUM(Sales[Revenue])
```

Conditional calculation

```
High Value Sales = IF(SUM(Sales[Revenue]) > 5000, "High", "Low")
```


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MEASURES AND CALCULATED COLUMNS

MEASURES

- A summarization of data
- Aggregate values from many rows: sums, averages, minimum, maximum, counts
- Results change as you interact with the report

CALCULATED COLUMNS

- An extension of a table
- A DAX formula evaluated for each row
- Adds a column that did not exist in the data source

```
Total Price = 'Sales OrderDetails'[Quantity] * 'Sales OrderDetails'[Unit Price]
```


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MAIN LAB · HANDS-ON

CREATE MEASURES

Add measures to the Contoso sales report. Three patterns to start from:

Count rows	`TotalRows = COUNTROWS(Sales)`
Count filtered rows	`HighValueSales = COUNTROWS(FILTER(Sales, Sales[Revenue] > 10000))`
Count distinct values	`UniqueCustomers = DISTINCTCOUNT(Sales[CustomerID])`

DAY 3 · CHAPTER 02 · VISUALIZATION

PRESENTING DATA GRAPHICALLY

Visualization helps users understand patterns, trends, relationships and insights quickly. In the main lab we go through three stages:

STAGE 1

CREATE THE REPORT

Add visuals that answer the main lab questions.

STAGE 2

DESIGN THE REPORT

Choose the right visual for each finding and format it.

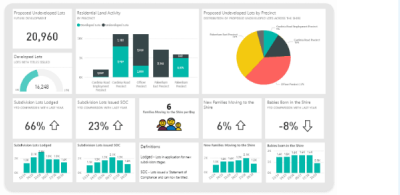
STAGE 3

DESIGN THE LAYOUT

Arrange the page so the story reads clearly.

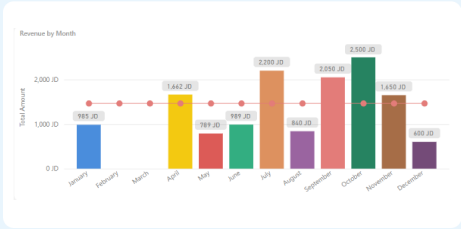
DESIGN A REPORT LAYOUT

The page layout depends on the business requirements, the context of the data and the output requirements.



DASHBOARD

High-level information on a single page.



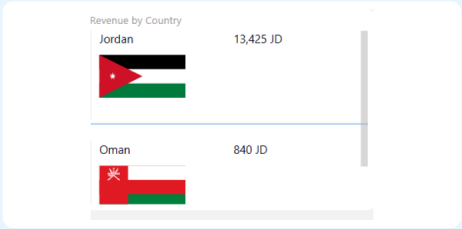
REPORT

A multi-perspective view, with visuals for different findings.



CONDITIONAL FORMATTING

Change colors, fonts or icons by rules and values to highlight trends and anomalies.



CUSTOM TOOLTIPS

Show extra context about a data point when users hover over it.

PUBLISHING A REPORT

Publishing and management means deploying, sharing, securing, refreshing and maintaining Power BI content after development.

1. Create the report in Power BI Desktop
2. Save the .pbix file
3. Select Publish
4. Sign in to Power BI
5. Choose a workspace
6. Click Publish



Desktop (.pbix) → Power BI service → your audience

ROW-LEVEL SECURITY (RLS)

Restricts access to data at the row level: different users see different subsets of the data, based on their roles.

STATIC METHOD

Uses a fixed value in the DAX filter.

One role per value, such as "Game" or "Clothing". Roles must be maintained by hand.

DYNAMIC METHOD

Uses a DAX function in the filter.

USERPRINCIPALNAME() is compared with the email in the Employees table. Set up once.

COURSE RECAP

5 STEPS. ONE WORKFLOW.

From raw data to real insight, on your own data next.

- ✓ **Prepare** with Power Query
- ✓ **Model** with relationships
- ✓ **Visualize** in reports
- ✓ **Analyze** with DAX
- ✓ **Manage** in the Power BI service

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QUESTIONS AND ANSWERS

THANK YOU FOR YOUR ATTENTION

Mohammad Sbeeh

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